

## S-2. Protection Wall /Embankment/Retaining Wall/Spurs



Forest / Shrub  
Land Conservation



## Medium



## High



Flood



Flood

 Flood Flood

## Flood



Flood



## Measure Type

## Cost

## Preferences by Community

## Hazards

## Implementation Time of Measures

## Descriptions

## Measures

A protective wall for hilly regions is a structural barrier that supports vulnerable hillside parts to avoid soil erosion, landslides, and slope destabilization. These walls, made of strong materials such as reinforced concrete, stone, or interlocking blocks, are strategically positioned along slopes to withstand the lateral pressure exerted by soil, rock, and water.

Landslides and soil erosion cause substantial problems to Bangladesh's CHT region, which is noted for its uneven topography and frequent excessive rainfall. Protecting walls, also known as retaining walls, are crucial adaptation structures in this region, helping to stabilize slopes, reduce erosion, and protect towns and infrastructure from landslide threats. Protective walls are an important adaptation measure in the CHT because they improve slope stability, reduce landslides, and protect communities and infrastructure. These structures increase resilience against climate change by addressing both the environmental and social issues of hillside stability.

### Location

|       |      |      |      |      |      |       |      |      |
|-------|------|------|------|------|------|-------|------|------|
| R-114 | R-99 | R-65 | K-98 | K-76 | K-71 | B-173 | B-77 | B-76 |
| ■     | ■    | ■    | ■    | ■    | ■    | ■     | ■    | ■    |

### Applicable Sites

- Agricultural Lands Adjacent to Water Bodies
- Slopes Prone to Soil Erosion
- Riparian Zones (Riverbanks & Stream Edges)
- Roadside and Settlement Perimeters

## Slope

| Less than 3% | 3% to 15% | 15% to 30% | 30% to 60% | Above 60% |
|--------------|-----------|------------|------------|-----------|
| √            | √         |            |            |           |

## Design options and performance

- Gravity Retaining Walls (Concrete or Stone Masonry) – Uses its own weight to resist soil pressure, best for low to medium slopes.
- Gabion Retaining Walls (Wire Mesh Filled with Rocks) – Flexible and cost-effective solution for erosion control.
- Reinforced Concrete Retaining Walls – Best suited for high-risk slopes requiring strong structural reinforcement.
- Crib Walls (Interlocking Concrete or Timber Frameworks) – Provides stability with vegetation integration for eco-friendly slope management.
- Terraced or Stepped Walls – Designed for very steep slopes by breaking up slope pressure into multiple levels.
- Soil-Nailed or Anchored Retaining Walls – Uses deep-anchored rods to reinforce loose soils on extreme slopes.

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|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|---------------|-------------|
| Feasibility criteria      | <p><b>Technical Feasibility:</b></p> <ul style="list-style-type: none"> <li>• Site-specific design assessment through geotechnical analysis.</li> <li>• Ensures load-bearing capacity and structural strength.</li> <li>• Integrates drainage solutions to prevent hydrostatic pressure.</li> </ul> <p><b>Environmental Feasibility:</b></p> <ul style="list-style-type: none"> <li>• Reduces soil erosion and mitigates landslides.</li> <li>• Manages water runoff to prevent flash floods.</li> <li>• Supports biodiversity and ecosystem preservation.</li> </ul> <p><b>Economic Feasibility:</b></p> <ul style="list-style-type: none"> <li>• Cost-effective solutions like gabion and gravity walls.</li> <li>• Provides long-term infrastructure protection.</li> <li>• Promotes local material use and labor involvement.</li> </ul> <p><b>Social Feasibility:</b></p> <ul style="list-style-type: none"> <li>• Enhances community safety and disaster risk reduction.</li> <li>• Generates employment for local workers.</li> <li>• Aligns with national disaster preparedness policies.</li> </ul> |            |               |             |
| Operation and maintenance | <ul style="list-style-type: none"> <li>• Regular Weeding &amp; Mulching – Maintains soil moisture and prevents invasive species growth.</li> <li>• Replanting &amp; Gap Filling – Ensures consistent buffer coverage and growth.</li> <li>• Drainage &amp; Water Flow Management – Maintains natural runoff channels to prevent waterlogging.</li> <li>• Community-Led Monitoring &amp; Capacity Building – Encourages local training on buffer strip maintenance.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |            |               |             |
| Cost                      | Medium                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |            |               |             |
| Benefits                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |            |               |             |
|                           | <b>Type</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Low</b> | <b>Medium</b> | <b>High</b> |
|                           | Canopy Cover Increase                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |            |               |             |
|                           | Carbon Sequestration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |            |               |             |
|                           | Crop Production Increase                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |            |               |             |
|                           | Crop Failure Risk Reduction                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |            |               |             |
|                           | Ensuring Food Security                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |            |               |             |
|                           | Fruit Production Increase                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |            |               |             |
|                           | Forestry Goods Production Increase                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |            |               |             |
|                           | Livestock Production Increase                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |            |               |             |
|                           | Fisheries Production Increase                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |            |               |             |
|                           | Alternative Livelihood Generation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |            |               |             |
|                           | Increase of Income                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |            |               |             |
|                           | Soil Erosion Control                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |            |               |             |
|                           | Improved Soil Health                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |            |               |             |
|                           | Soil Fertility Increase                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |            |               |             |
|                           | Sustainable Water Supply                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |            |               |             |
|                           | Water Quality Improvement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |            |               |             |

|                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |            |               |             |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|---------------|-------------|
|                                  | Water Purification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |            |               |             |
|                                  | River Bank Erosion Control                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |            |               |             |
|                                  | Surface Runoff Reduction                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |            |               |             |
|                                  | Groundwater Recharge                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |            |               |             |
|                                  | Pest and Disease Management                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |            |               |             |
|                                  | Reduction of Pesticides and Fertilizer Use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |            |               |             |
|                                  | Ecosystem and Biodiversity Increase                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |            |               |             |
|                                  | Better Communications                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |            |               |             |
|                                  | Better Education                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |            |               |             |
|                                  | Better WASH facilities                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |            |               |             |
| <b>Implementation Challenges</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |            |               |             |
|                                  | <b>Type</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Low</b> | <b>Medium</b> | <b>High</b> |
|                                  | Land unavailable/high land area required                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |            |               |             |
|                                  | Supply chain disconnection                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |            |               |             |
|                                  | Land right related problems                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |            |               |             |
|                                  | High human investment capital need                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |            |               |             |
|                                  | Insufficient infrastructures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |            |               |             |
|                                  | Less demand of agroforestry product                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |            |               |             |
|                                  | Environmentally unsustainable                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |            |               |             |
|                                  | Socially unacceptable                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |            |               |             |
|                                  | Less beneficial                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |            |               |             |
| <b>Environmental performance</b> | <ul style="list-style-type: none"> <li>• Landslide &amp; Soil Erosion Prevention – Stabilizes hillsides and prevents mass movements.</li> <li>• Enhanced Water Retention &amp; Groundwater Recharge – Improves hydrological balance by controlling runoff.</li> <li>• Carbon Sequestration Potential (With Green Infrastructure) – Walls integrated with vegetation contribute to ecosystem restoration.</li> <li>• Reduction in Infrastructure Damage – Protects roads, bridges, and settlements from climate-induced hazards.</li> </ul> |            |               |             |
| <b>Remarks</b>                   | This measure will aid landscape restoration in CHT of Bangladesh along with creating alternative livelihood opportunities.                                                                                                                                                                                                                                                                                                                                                                                                                 |            |               |             |